lka_controller.c

```
#include <stdio.h>

#define DELTA_T 0.01f
#define SAT_MAX_ANGLE 0.60f
#define SAT_MIN_ANGLE -0.60f

static const float Kp = 0.1800f;
static const float Ki = 0.0100f;
static const float Kd = 0.0500f;

static float erreur_precedente = 0.0f;
static float terme_integral = 0.0f;

float LKA_ComputeSteering(float consigne_y, float mesure_y) {
    float erreur;
    float terme_proportionnel;
    float terme_derive;
    float commande_brute;
    float commande_saturee;

    erreur = consigne_y - mesure_y;
    terme_proportionnel = Kp * erreur;
    terme_integral += erreur * DELTA_T;
    terme_derive = (erreur - erreur_precedente) / DELTA_T;

    commande_brute = terme_proportionnel + (Ki * terme_integral) + (Kd * terme_derive);

    if (commande_brute > SAT_MAX_ANGLE) {
        commande_saturee = SAT_MAX_ANGLE;
    } else if (commande_brute < SAT_MIN_ANGLE) {
        commande_saturee = SAT_MIN_ANGLE;
```

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    } else {
        commande_saturee = commande_brute;
    }

    erreur_precedente = erreur;
    return commande_saturee;
}
```